

Correction Note — Energy Drift Analysis (Step 5)

This document outlines corrections and clarifications to the original 'Energy Drift Tracking — SIMON vs ias15' report. These updates improve scientific precision, remove inconsistencies, and align conclusions with the experimental evidence.

1. Clarification of Energy Drift Source

Original claim: Energy drift is entirely caused by the leapfrog integrator.

Correction: Energy drift is dominated by the leapfrog/adaptive integration scheme. For IC3 and IC4, the NN is never invoked ($NN_frac = 0.00$), confirming the drift originates from the integrator. For IC1 ($NN_frac = 0.45\%$), the NN may contribute minimally, but is not the dominant source of error.

2. IC4 Drift Interpretation

Original phrasing suggested comparable drift across configurations.

Correction: IC4 exhibits significantly higher fractional drift (13.77%) compared to IC1 (1.39%) and IC3 (0.67%). This is due to its weakly bound hierarchical structure and small initial energy $|E_0|$, which amplifies fractional drift.

3. Summary Table Correction

The statement 'SIMON drift < 1% in all cases' is incorrect.

Correct values:

- IC1: 1.39%
- IC3: 0.67%
- IC4: 13.77%

This line should be removed or replaced with an accurate range description.

4. Removal of Ratio Interpretation

The SIMON/ias15 energy drift ratio ($\sim 10^{12}$ – 10^{14}) is mathematically correct but scientifically misleading. It reflects differences in integrator order rather than model quality. This metric should not be included in the main report or figures.

5. Notation Fixes

Replace corrupted symbols with standard notation:

- E_0 instead of E■
- 10^{-14} instead of 10^{14} ■
- $\Delta E/E_0$ for fractional drift
- M_{sun} for solar mass

6. Final Revised Conclusion

SIMON exhibits 0.67–1.39% energy drift over 100 years for primary configurations, with a higher value (13.77%) in a weakly bound hierarchical case. Analysis of zero-NN-invocation scenarios confirms that energy drift is primarily driven by the leapfrog/adaptive integration scheme rather than the neural network correction.